

Oxidation Numbers and Redox Reactions

Answer key with worked solutions

A. Understanding

1. Definitions.

Oxidation is the loss of electrons; the oxidation number **increases**. **Reduction** is the gain of electrons; the oxidation number **decreases**.

B. Oxidation numbers

2. Oxidation numbers.

(a) **-3** (b) **+5** (c) **+6** (d) **+6** (e) **+3** (f) **-1** (peroxide) (g) **+4** (h) **+5**. For example, (h): $x + 3(-2) = -1$, so $x = +5$.

C. Agents and half-equations

3. Chlorine and bromide.

Br⁻ is oxidized ($-1 \rightarrow 0$) and **Cl₂ is reduced** ($0 \rightarrow -1$). **Oxidizing agent: Cl₂; reducing agent: Br⁻.**

4. Half-equations.

Oxidation: $\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$. Reduction: $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$.

5. Redox or not?

(a) **Not redox**: no oxidation number changes. (b) **Redox** (disproportionation): O goes from -1 to -2 in H₂O and to 0 in O₂. (c) **Not redox**: a precipitation in which all oxidation numbers stay the same.

D. Balancing

6. Copper and silver.

Cu loses $2 e^-$; each Ag^+ gains $1 e^-$, so 2 Ag^+ are needed: $Cu + 2 Ag^+ \longrightarrow Cu^{2+} + 2 Ag$.
Charge: $+2 = +2$. ✓

7. Dichromate and iron(II).

Reduction: $Cr_2O_7^{2-} + 14 H^+ + 6 e^- \longrightarrow 2 Cr^{3+} + 7 H_2O$

Oxidation ($\times 6$): $6 Fe^{2+} \longrightarrow 6 Fe^{3+} + 6 e^-$

Overall: $Cr_2O_7^{2-} + 14 H^+ + 6 Fe^{2+} \longrightarrow 2 Cr^{3+} + 7 H_2O + 6 Fe^{3+}$. Charge: $-2 + 14 + 12 = +24$ on the left; $+6 + 18 = +24$ on the right. ✓

8. Copper and nitrate.

Oxidation ($\times 3$): $3 Cu \longrightarrow 3 Cu^{2+} + 6 e^-$

Reduction ($\times 2$): $2 NO_3^- + 8 H^+ + 6 e^- \longrightarrow 2 NO + 4 H_2O$

Overall: $3 Cu + 2 NO_3^- + 8 H^+ \longrightarrow 3 Cu^{2+} + 2 NO + 4 H_2O$. Charge: $-2 + 8 = +6 = +6$. ✓

E. Challenge

9. Permanganate.

(a) Reduction ($\times 2$): $2 MnO_4^- + 16 H^+ + 10 e^- \longrightarrow 2 Mn^{2+} + 8 H_2O$; oxidation ($\times 5$):
 $5 C_2O_4^{2-} \longrightarrow 10 CO_2 + 10 e^-$

Overall: $2 MnO_4^- + 5 C_2O_4^{2-} + 16 H^+ \longrightarrow 2 Mn^{2+} + 10 CO_2 + 8 H_2O$. Charge: $-2 - 10 + 16 = +4 = +4$. ✓

(b) $n(MnO_4^-) = 0.0200 \text{ mol/L} \times 0.02000 \text{ L} = 4.00 \times 10^{-4} \text{ mol}$

$n(Fe^{2+}) = 4.00 \times 10^{-4} \text{ mol } MnO_4^- \times \frac{5 \text{ mol } Fe^{2+}}{1 \text{ mol } MnO_4^-} = 2.00 \times 10^{-3} \text{ mol}$

$[Fe^{2+}] = \frac{2.00 \times 10^{-3} \text{ mol}}{0.02500 \text{ L}} = 0.0800 \text{ M}$

10. Basic solution.

Oxidation (in acid, $\times 2$): $2 \text{Cr}(\text{OH})_3 + 2 \text{H}_2\text{O} \longrightarrow 2 \text{CrO}_4^{2-} + 10 \text{H}^+ + 6 \text{e}^-$

Reduction (in acid, $\times 3$): $3 \text{ClO}^- + 6 \text{H}^+ + 6 \text{e}^- \longrightarrow 3 \text{Cl}^- + 3 \text{H}_2\text{O}$

Add and cancel: $2 \text{Cr}(\text{OH})_3 + 3 \text{ClO}^- \longrightarrow 2 \text{CrO}_4^{2-} + 3 \text{Cl}^- + 4 \text{H}^+ + \text{H}_2\text{O}$

Add 4OH^- to both sides ($4 \text{H}^+ + 4 \text{OH}^- \longrightarrow 4 \text{H}_2\text{O}$):

$2 \text{Cr}(\text{OH})_3 + 3 \text{ClO}^- + 4 \text{OH}^- \longrightarrow 2 \text{CrO}_4^{2-} + 3 \text{Cl}^- + 5 \text{H}_2\text{O}$

Check: O: $6 + 3 + 4 = 13$ and $8 + 5 = 13$; H: $6 + 4 = 10$ and 10 ; charge: $-3 - 4 = -7$ and $-4 - 3 = -7$. ✓